

Temozolomide non-completion and absence of residual enhancement are associated with distant recurrence in IDH-wildtype glioblastoma

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OBJECTIVES Glioblastoma (GBM) recurrence pattern determines subsequent management: local recurrence may favour re-resection or re-irradiation, while distant recurrence directs systemic therapy and trial enrolment. Clinicians lack validated tools to anticipate recurrence pattern using early post-operative and treatment-course variables. We used a retrospective single-centre registry to identify early predictors of local versus distant recurrence.

METHODS Patients undergoing surgery for histologically confirmed GBM at Queen's Medical Centre, Nottingham between 2017 and 2023 were identified. Patients with IDH mutation or unavailable IDH status were excluded. Recurrence pattern at first progression was classified as local (contiguous with the original resection cavity) or distant on follow-up imaging. A two-predictor multivariable logistic regression model was constructed, limited to two predictors given the number of distant-recurrence events to minimise overfitting, with complete-case analysis applied for missing predictor data. Discrimination was assessed by area under the receiver operating characteristic curve (AUC).

RESULTS Sixty-three IDH-wildtype patients with confirmed recurrence classification were included (local n=45, distant n=18). On multivariable analysis (complete cases, n=55), temozolomide non-completion (OR 12.4, 95% CI 1.7–88.7, p=0.012) and absence of residual contrast enhancement on early post-operative MRI (OR 4.9, 95% CI 1.2–20.4, p=0.030) were associated with distant recurrence (AUC 0.773).

CONCLUSIONS Absence of residual enhancement, a routinely captured early imaging marker of local disease control, was associated with distant recurrence. Temozolomide non-completion may reflect treatment intolerance or early occult progression, limiting causal interpretation. Both variables are routinely available before first progression and define a clinically tractable window for early risk stratification. Effect estimates were imprecise with wide confidence intervals given the limited number of distant events; ongoing registry expansion will enable validation and translation into a clinical decision tool.